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# The positive influence of Tibetan Buddhist monasteries on forest conservation

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## Abstract

The Qinghai-Tibetan Plateau (QTP) is vital for stabilizing climate in East and South Asia and its glaciers and snow meltwater nourish key continental rivers. The region's forests are not only ecologically essential but also fragile. The practices of local herdsmen, shaped by long-standing cultural traditions, significantly affect forest dynamics. This study examines the vital role that Tibetan Buddhist monasteries play in forest conservation on the QTP. We utilize data sources including extensive field surveys and geospatial data to assess the impact of monastery and village locations on forest conditions. Spatial analysis techniques, such as Moran's I and Ripley's K function, reveal significant clustering of monasteries and villages, which are closely linked with forest conditions. Our findings indicate that forests near Buddhist monasteries have higher aboveground biomass and better quality compared to other areas, suggesting that their presence supports better forest conservation practices. This positive influence is attributed to the cultural significance of these sites and the environmental consciousness promoted by Buddhist cultural teachings. Additionally, this study employs analytical methods including Random Forest and Partial Least Squares Structural Equation Modeling (PLS-SEM) to explore the drivers of forest quality. Initial aboveground biomass, climatic factors, and monastery density emerge as key influences on forest quality within monastery buffer zones, indicating that cultural factors are crucial in shaping forest landscapes.


**Keywords** Forest condition, Buddhist monasteries, Land use, Random Forest, PLS-SEM

## Introduction

Forests play a crucial role in global environmental change and carbon cycling, significantly contributing to climate regulation [1, 2]. They absorb about a quarter of anthropogenic carbon emissions, acting as carbon sinks by storing carbon in tree biomass and soil. Additionally, they regulate the exchange of energy and water between the soil and the atmosphere [3]. Due to the unique terrain and climate of the Qinghai-Tibet Plateau (QTP), forests in this region play a critical role in regulating the regional climate, maintaining soil and water, and supporting other vital ecological functions [4]. However, the inherent fragility of these ecosystems makes them particularly vulnerable to external disturbances [5]. Recent research has identified significant threats to these forests, including population growth, intensified grazing, and the rapid expansion of the tourism industry [6, 7]. Moreover, large-scale infrastructure projects, such as the development of transportation corridors, have further contributed to forest degradation [8]. Human activities have not only accelerated deforestation but also led to severe forest degradation, posing significant threats to biodiversity and ecosystem services, with long-term negative impacts on local and global environmental conditions [9]. Therefore, urgent attention should be paid to the status of the QTP's forest ecosystems [10, 11].

Within the System of Environmental-Economic Accounting Ecosystem Accounting (SEEA-EA) framework, the extent and condition of forests are crucial

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indicators [12]. Forest extent primarily encompasses changes in forest area, with studies examining forest area at both global and local scales [13]. Forest condition focuses more on the physical, chemical, and biological characteristics of forests [14]. Among these, aboveground biomass is regarded as a relatively stable assessment indicator. Furthermore, scholars are increasingly inclined to consider the direct or indirect benefits that forest ecosystems provide to humans, widely known as ecosystem services [15]. Current research tends to concentrate on changes in forest area, forest condition, and the ecosystem services thereby provided, and various factors such as human activities and climate change that influence forest ecosystems [16]. However, one aspect that has received less attention is the role of cultural and religious influences on ecosystems. In particular, the impact of culture and religion on ecosystems through their influence on human behavior is an area that warrants deeper exploration [17]. For example, some studies demonstrate the positive effects of traditional farming practices in promoting the sustainability of agricultural ecosystems in traditional villages [18]. Others quantify the protection of forests through the lens of social perception of traditional knowledge [19]. Recognizing these cultural dimensions is essential for developing a comprehensive understanding of ecosystem conservation, especially in regions where traditional beliefs are deeply intertwined with daily life.

Given this context, Tibetan Buddhism—introduced to the QTP in the mid-seventh century CE—offers a unique perspective on how culture influences ecosystem conservation. Central to Buddhist philosophy is the concept of interdependence or interconnectedness, which posits that all phenomena, including living beings and the natural world, are intricately connected and mutually dependent [20]. This perspective instills a profound sense of reverence and care for the environment. Moreover, Tibetan Buddhism promotes an ethic of non-harming and compassion towards sentient life [21]. This ethical stance is encapsulated in the principle of "ahimsa," advocating the avoidance of harm to oneself and others, including animals and the environment. The practice of mindfulness, a central tenet of Buddhist meditation, further cultivates a deep awareness and appreciation of the natural world [22]. Practitioners are encouraged to be fully present and attentive to their surroundings, fostering a profound connection with nature [23].

Monasteries often function as centers for promoting environmental awareness and nature conservation [24]. Monastic communities engage in practical conservation activities, including tree planting and enforcing prohibitions against logging and hunting within sacred boundaries [19]. These religious sites are revered as sacred spaces, with surrounding forests integral to their spiritual significance [25]. Furthermore, monasteries play a critical role in educating local communities about environmental stewardship. Through sermons and communal activities, monks instill values of respect and harmony with nature, influencing local attitudes and behaviors toward forest conservation [26, 27]. In addition to their influential role in environmental education and practical conservation efforts, monasteries further contribute to forest conservation by mediating governance challenges arising from ambiguous land tenure and control among local communities [28]. In this context, the unofficial participation of monasteries becomes indispensable. Monasteries act as mediators and entities capable of mobilizing community action and bridging governance gaps. Their respected status within the community enables them to facilitate effective forest management practices, making their involvement crucial for conservation initiatives [29]. Most existing research focuses on the cultural and spiritual aspects without assessing tangible conservation outcomes [30, 31]. Therefore, it is imperative to quantitatively assess how Buddhist monasteries influence forest aboveground biomass and whether they effectively facilitate forest conservation on the QTP.

Changdu City, located in the eastern part of the Tibet Autonomous Region, is one of the originating locations of the Tibetan literary masterpiece, Epic of King Gesar. The Gesar culture has been preserved through various mediums, including oral legends, historical relics, murals and artifacts in Buddhist monasteries, as well as local customs and festivals [29]. The area is also characterized by a relatively concentrated forest distribution, reflecting the deep connection between cultural traditions and the natural landscape. Understanding the state of forests in Changdu from 2000 to 2020 and the impact of monasteries and villages on forests can inform targeted conservation strategies for cultural and ecological preservation. The main objectives of this study are as follows: (1) Quantify the distributional and associative characteristics of monastery and village sites in Changdu; (2) Quantify the changes in forest area and quality; (3) Identify the specific drivers of forest quality and elucidate the pathways through which monasteries influence forest conservation.

## Material and methods

### Study area

Changdu is located in the southeastern part of China (93° 50′ E–99° 10′ E, 28° 20′ N–33° 30′ N) (Fig. 1), at the transition between the Qinghai-Tibet Plateau and the high mountain valleys of southeastern Tibet [32]. It lies in the catchment areas of the Hengduan Mountains and three major rivers—Jinsha, Lancang, and Nu—which flow through the region, creating a unique landscape of

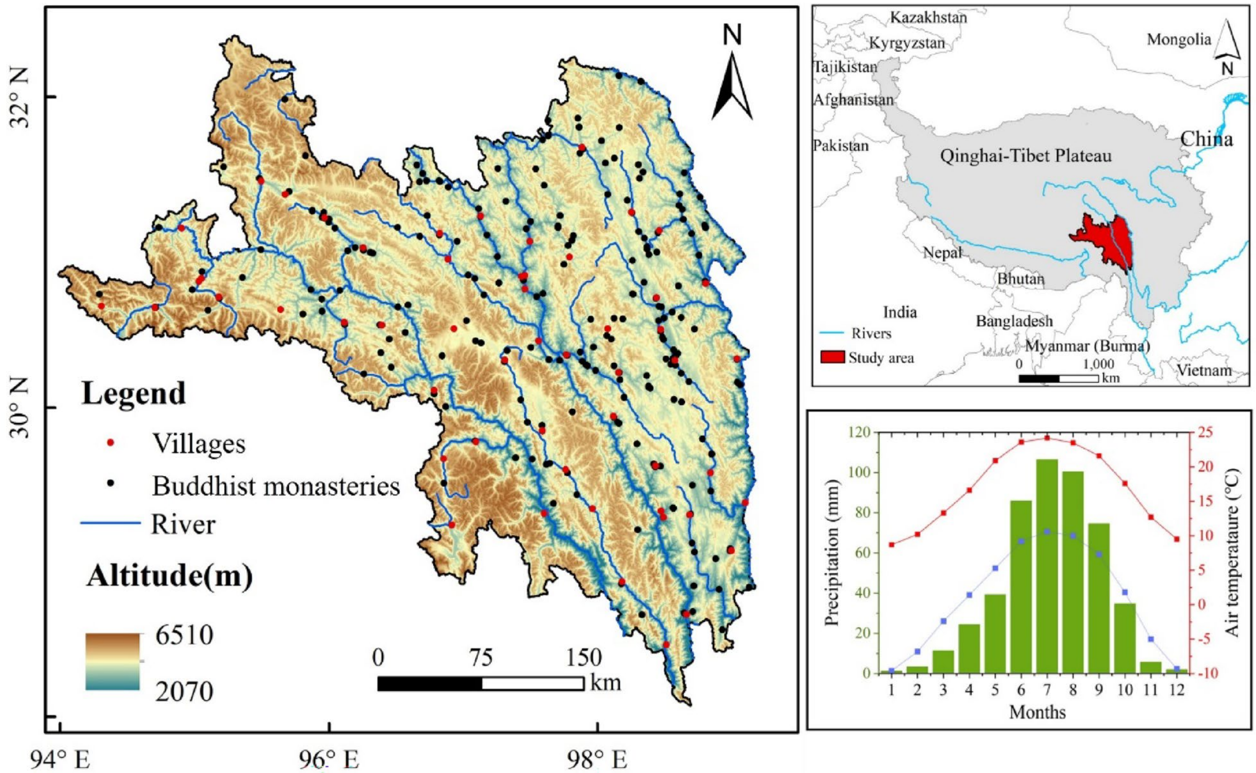


**Fig. 1** Distribution of monasteries and villages in Changdu and their geo-climatic characteristics

high mountains, deep valleys, and alternating ridges [33]. The climate is a plateau-subtropical monsoon type, with average annual temperatures ranging from 2.4 to 16.6 °C, and most of the annual 477.7 mm precipitation falls between May and September [34]. Vegetation is distributed in clear vertical zones: temperate grasslands below 3900 m, forests between 3900 and 4400 m, and alpine grasslands above 4400 m [35]. Additionally, Changdu is a significant religious center for Tibetan Buddhism, notably associated with the Epic of King Gesar. Buddhist monasteries serve as key religious, educational [36], and cultural institutions, housing murals, artifacts, and hosting significant religious festivals. These monasteries are not only spiritual centers but also places of philosophical education, attracting monks, pilgrims, and visitors for religious debates and other spiritual activities [37]. The religious influence also extends to environmental conservation, with many sacred forests protected by religious customs [38].

## Data sources

This study utilizes data from diverse sources, covering various environmental and human activity aspects. Monastery data was gathered through extensive field surveys from August 2, 2011, to August 20, 2018. The surveys were primarily conducted during the forest growing season and the Tibetan New Year, recording spatial locations of confirmed monasteries using GPS devices and conducting interviews with local abbots and monks. Village data was sourced from the National Bureau of Statistics (https://www.stats.gov.cn/). Environmental data, including aboveground biomass and grazing intensity, with a spatial resolution of 1 km, were obtained from the National Tibetan Plateau Data Center (https://data.tpdc.ac.cn) for the period from 1990 to 2020. Soil data for the same period was retrieved from the World Soil Database (https://www.fao.org), while climate data from 1990 to 2020 were sourced from the Resource Environment Science and Data Platform (www.resdc.cn). Geographical data, including topographic data with a spatial resolution of 12.5 m, was sourced from the National Tibetan Plateau Data Center, and land cover/land use data with a spatial resolution of 30 m, covering the period from 1990 to 2020, were obtained from the China Land Cover Database (www.zendo.org).

## Methods

### *Analysis of point patterns*

In order to investigate whether there are features such as spatial clustering and spatial outliers in the distribution of monastery and village locations, this study employs Exploratory Spatial Data Analysis (ESDA):

$$Moran'sI = \frac{n\sum_{i=1}^{n}\sum_{j=1}^{n} W_{ij}(X_i - \bar{X})(X_j - \bar{X})}{\sum_{i=1}^{n}\sum_{j=i}^{n} W_{ij}\sum_{i=1}^{n}(X_i - \bar{X})^2} \tag{1}$$

where n is the observed value of the study object; $x_i$ and xj are the observed values; x is the mean of $x_i$; $W_{ij}$ is the spatial weight of study object i and object j, the connection matrix (i, j = 1, 2, 3, …, n). The global *Moran's I* takes values generally between −1 and 1. If it is greater than 0, it means that there is positive autocorrelation in the space; if it is less than 0, it means the spatial entities are discrete; if it tends to 0, it means that if it is less than 0, the spatial entities are discrete; if it is close to 0, the spatial entities are randomly distributed [39, 40].

#### *Spatial correlation analysis of monasteries and villages*

To investigate the spatial relationship between monastery and village locations, we employed bivariate Ripley's K(r) function and bivariate pair correlation function as analytical tools. These functions are capable of assessing the degree of spatial correlation between point patterns, providing deeper insights into the relationship between the spatial distributions of monasteries and villages [41].

Ripley's *K(r)* function is a commonly used method for assessing spatial correlation between point patterns. It measures the average distance between pairs of points within a specified distance $r$, and compares it with that of a random point distribution:

$$K(r) = \frac{1}{n^2\lambda}\sum_{i=1}^{n}\sum_{j=1, j\neq i}^{n}\frac{I(||x_i - x_j|| \leq r)}{n} \tag{2}$$

where $n$ is the total number of points, $\lambda$ is the point density, $x_i$ and $x_j$ are the coordinates of the $i$th and $j$th points respectively, and $I$ is the indicator function, which equals 1 if its argument is true and 0 otherwise. These functions are compared against random models or simulated data to ascertain significant spatial correlation. We set the threshold at a 95% confidence level. If observed function values surpass theoretical values, this indicates spatial aggregation between monasteries and villages. Conversely, values below suggest dispersion. If falling within the confidence interval, it implies random distribution of the two entities.

The pair correlation function is utilized to analyze the correlation between monastery and village locations across various distance ranges. It offers insights into the pairwise relationships between points, aiding in understanding their associations at different scales. Its formula is given by:

$$C(r) = \frac{1}{n(n-1)}\sum_{i=1}^{n}\sum_{j=1, j\neq i}^{n}\delta(||x_i - x_j|| - r) \tag{3}$$

where $\delta$ is the Dirac delta function, which evaluates whether the distance between pairs of points equals $r$, yielding 1 if true and 0 otherwise. If the $C(r)$ n is greater than 1 for a certain distance r, it indicates that points are more likely to be found at this distance from each other than would be expected under a completely random distribution (indicating clustering). If $C(r)$ is less than 1, it suggests that points are less likely to be found at this distance from each other than would be expected under a completely random distribution (indicating dispersion). If $C(r) = 1$, it suggests a completely random distribution.

#### *Buffer analysis of forest quality*

Buffer analysis, a method of creating a zone around a geographical point based on a specific range, is used to study features, resource distribution, and potential risks. The impact increases with proximity to the central point. We used this technique to evaluate the environmental impact of specific points like monasteries and villages [16, 42]. We created 6 km radius buffers around these points using ArcGIS10.8, guided by academic literature and the plateau's socio-ecological conditions. We analyzed variables within these buffers, focusing on forest area and quality as key environmental indicators. Our study, covering 237 monastery and 87 village points, offers a local landscape perspective.

#### *Random forest algorithm driver analysis*

To investigate the drivers of the distribution of monasteries and villages, we conducted a random forest regression analysis. We categorized these factors into four major classes: climate factors (including precipitation—PRE, and temperature—TEM), natural environment factors (initial aboveground biomass—IABG, slope, digital elevation model—DEM, soil organic carbon—SOC, and soil total nitrogen—SN), human Activity Index (population—POP and grazing intensity), and monasteries density. Through random forest regression analysis, we aimed to uncover the influence of these factors on the distribution of monasteries and villages, and compare the differences in the importance of factors across different buffer zones [41]. Additionally, population, grazing intensity, precipitation, temperature, aboveground biomass, and soil attributes were measured from 1990 to 2020. We used the formula:

$$y_i = f(x_i) + \varepsilon_i \tag{4}$$

where $y_i$ represents the response variable (monasteries and villages) at location $i$; $x_i$ represents the vector of predictor variables; $\varepsilon_i$ represents the error term. The model aims to estimate the relationship $f$ by growing an ensemble of decision trees. The importance of each predictor variable $x_i$ can be assessed based on metrics such as mean decrease in impurity, mean decrease in accuracy, or mean decrease in node impurity, which quantify the contribution of each variable to the overall predictive performance of the model.

### *Partial least squares-structural equation modelling (PLS-SEM)*

Partial Least Squares Structural Equation Modeling (PLS-SEM) is a statistical technique that combines elements of both structural equation modeling (SEM) and principal component analysis. PLS-SEM is particularly useful when investigating complex relationships among latent variables and observed indicators in situations where the data may not adhere strictly to multivariate normality or sample sizes are relatively small [43, 44].

In this paper, the application of the PLS-SEM model aims to elucidate the interconnections and interplays among distinct latent variables. Through the incorporation of observed indicators designed to quantify these latent constructs, the model facilitates the discernment of direct and mediated effects within this network of variables. This analytical approach offers valuable insights into the pivotal factors that wield significant influence over the forest status, contributing to a more comprehensive understanding of the intricate relationships among variables. The measurement equation is:

$$Y = \Lambda_y \alpha + \beta \tag{5}$$

$$X = \Lambda_x \lambda + \varepsilon \tag{6}$$

where $X$ and $Y$ represent latent variables and observed variables respectively; $\Lambda$ refers to the relationship between latent variable and corresponding observed variable; $\alpha$ and β represent latent variables and observed variables respectively; $\varepsilon$ represent the measurement error.

The structural equation is then:

$$\alpha = B\alpha + \Gamma\xi + \lambda \tag{7}$$

where λ represents a vector of exogenous latent variables; α represents a vector of endogenous latent variables; $B$ represents the relationships among endogenous latent variables α; Γ represents the impact of exogenous latent variables $\lambda$ on endogenous latent variables α.

## Results

### Spatial distribution of monastery and village points

The spatial distribution of point features is usually assessed using Moran's $I$ to determine whether it is clustered, random, or dispersed (Fig. 2). In this study, the nearest neighbor indices of monasteries and villages points were calculated using ArcGIS 10.8, and the average actual nearest neighbor distances were found to be 33.26 km and 57.32 km for the two types of points, respectively, while the theoretical nearest neighbor distances were 55.49 km and 72.63 km, respectively; the nearest neighbor ratio was 0.70. The global Moran's I test was used to examine the spatial distribution of monastery and village points in the study area, and the Z values were respectively −10.95 and −10.59, which was significantly less than the critical value of −2.58 at the significance level of 1%, indicating spatial clustering pattern of monastery and village points. In our field investigation, we observed that monasteries are predominantly situated

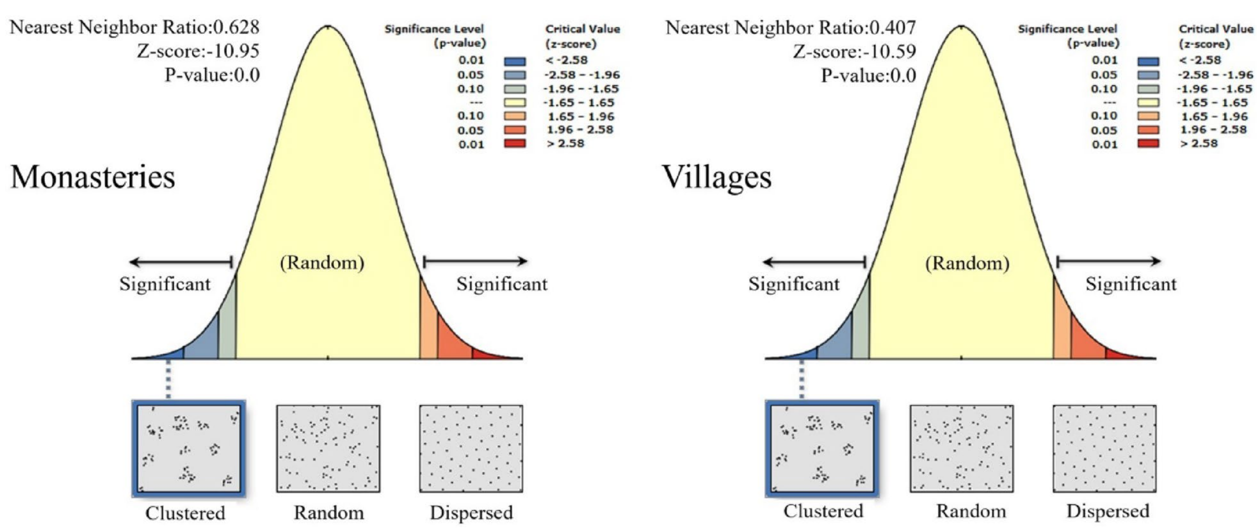


**Fig. 2** Distributional characteristics of monasteries and villages

at higher elevations. Statistical analysis indicated that the average elevation of monastery sites is higher than that of villages, but this difference is not statistically significant (Fig. 3). On the other hand, monastery sites are significantly more common in regions with steeper slopes. This suggests that sacred mountains, which are prominent in Tibetan Buddhism, may potentially influence the distribution of monastery locations.

The observed bivariate Ripley's $K(r)$ values surpass the expected values and fall within the confidence interval, indicating a significant spatial clustering of monastery and village distributions (Fig. 4). Particularly noteworthy is the consistent elevation of observed Ripley's $K(r)$ values, exceeding the theoretical Ripley's $K(r)$ values, especially evident as distances increase to 20 km. This suggests a pronounced spatial association between monasteries and villages across varying distances, extending up to 20 km. This deviation of observed values from the theoretical values underscores the presence of non-random spatial patterns, indicative of a structured relationship between monastery and village distributions. The pair correlation function analysis reveals distinct spatial relationships among monasteries and villages, as well as among monastery points and village points. Both monasteries and villages exhibit significant clustering tendencies, indicating a strong spatial interaction between them. Specifically, the former demonstrate a clustering pattern within the 0–5 km range, suggesting a tendency for monasteries to cluster together at shorter distances, possibly indicative of religious or cultural centers. Conversely, among village points, there is an initial increasing trend followed by a diminishing trend within the 0–20 km range, reflecting a significant clustering effect at shorter distances that gradually weakens with increasing distance.

## Extent and quality changes in forest landscapes

### *Change of forests area*

Within the spatial extent of Changdu, forest area accounted for 15.4% of total area in 2020. From 1990 to

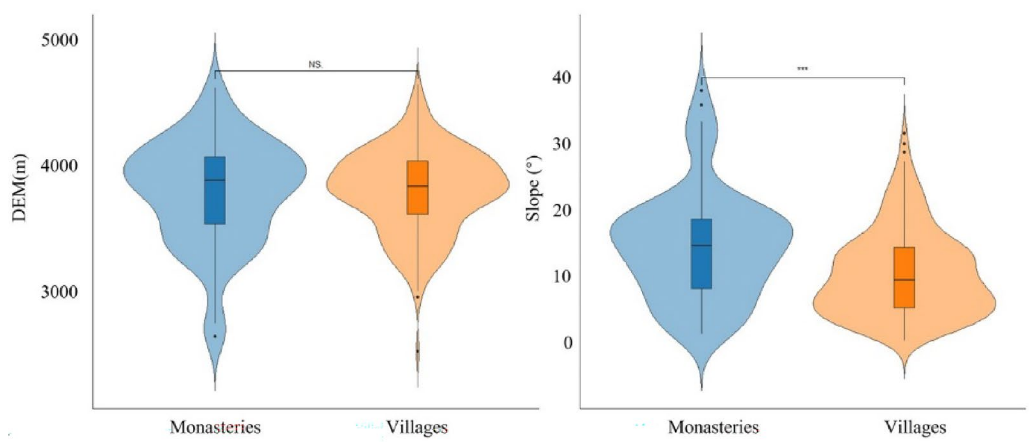


**Fig. 3** Terrain differences in the distribution of monasteries and villages

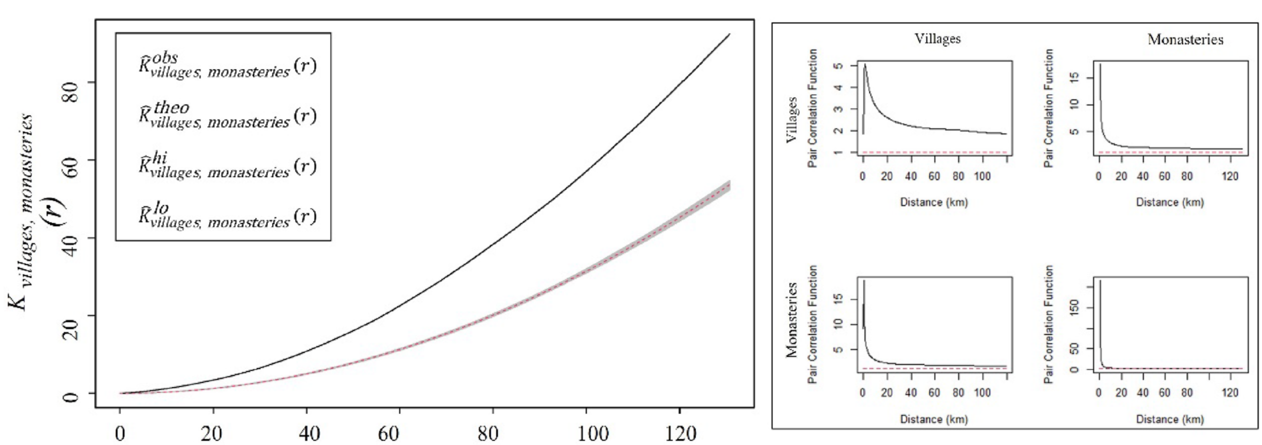


**Fig. 4** Correlation characteristics of monasteries and villages

2020, there was a continuous increasing trend in forest cover, although the rate of growth gradually declined. Changes in land use within the region spanned two distinct periods, as illustrated in Fig. 5. Initially, from 1990 to 2010, the most significant changes occurred in the southwestern part of the region. Subsequently, from 2010 to 2020, the overall magnitude of land use changes decreased, with more pronounced changes in the northeastern part of the region, primarily involving transformations between forest and grassland.

Between 1990 and 2020, we analyzed the changes in forest area within the buffer zones surrounding monastery and village sites. The findings indicated that the forest area within monastery buffer zones increased by 16.9%, corresponding to an expansion of 3841 $km^2$. This rate of increase was greater than the 16.4% growth observed within the village buffer zones. To evaluate the statistical distribution of the forest change data between these buffer zones, we utilized both a Quantile–Quantile plot (Q–Q plot) and the Shapiro–Wilk test (Fig. 6). The analysis from the Q–Q plot, alongside the Shapiro–Wilk test results, with a W value of 0.88645 and a $p$-value greater than 0.05, collectively indicate that the data do not follow a normal distribution. Consequently, we proceeded with the Mann–Whitney U test to analyze the differences further. The Mann–Whitney U test results indicated significant differences in forest changes between the monastery and village buffer zones in Changdu from 1990 to 2020. Specifically, the increase in forest area within monastery buffer zones was significantly higher than that in the village buffer zones. This analysis highlights a distinct pattern of greater forest expansion in areas surrounding monasteries compared to those surrounding villages.

### *Change of forest quality*

Over the study period, Changdu's aboveground biomass (ABG) values displayed a consistent upward trajectory (Fig. 7), rising from 98.27 t/ha in 1990 to 273.29 t/ha in 2020, indicative of significant improvements in vegetation cover and ecological health. Comparatively, monastery consistently exhibited higher ABG values than villages, implying a more robust ecosystem around these cultural sites. This pattern suggests that the conservation efforts within monastery vicinities nurtured better conditions for the natural environment,

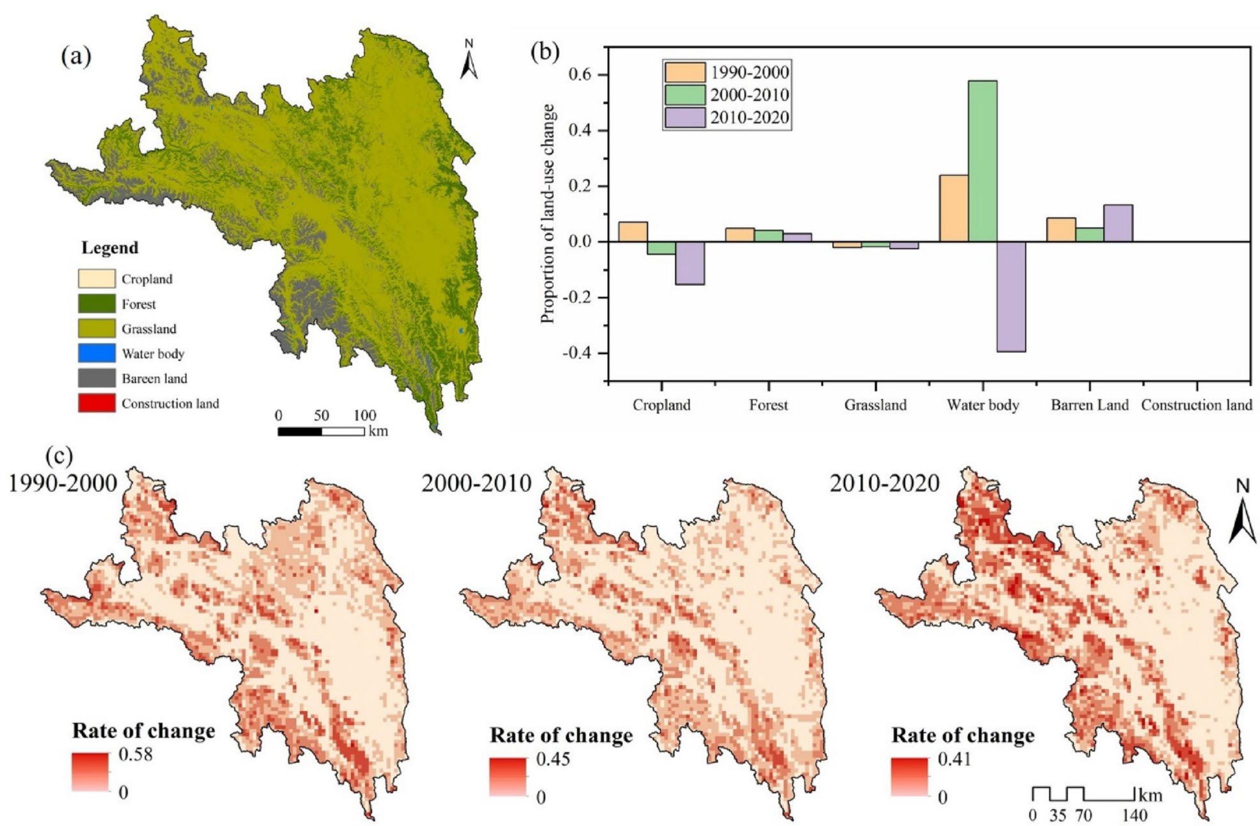


**Fig. 5** Land use change in Changdu from 2000 to 2020. **a** Represents the spatial distribution of land use in Changdu in 2020, **b** represents land use change in Changdu from 1990 to 2020, and **c** represents the spatial change in land use, 1990–2020

reflecting a harmonious interplay between cultural practices and ecological stewardship.

ABG had significant differences between monastery and village buffer zones (Fig. 8). Aboveground biomass showed an increasing tendency with distance from the buffer zones. From 1990 to 2020, increases around monasteries were more pronounced, but the rate decreased after 2000. Overall, above-ground biomass was significantly higher within monastery buffer zones.

#### Results of driver factor analysis

To investigate the factors influencing the quality of forests surrounding monasteries and villages, we employed the random forest algorithm to identify the primary drivers. Our motivation stemmed from the need to better understand the dynamics shaping forest ecosystems in proximity to cultural and residential sites. Analysis of the training set revealed robust fitting results, with adjusted R-squared values of 0.95 within monastery buffer zones and 0.94 within village buffer zones (Fig. 9). In the testing phase, the model's performance within monastery buffer

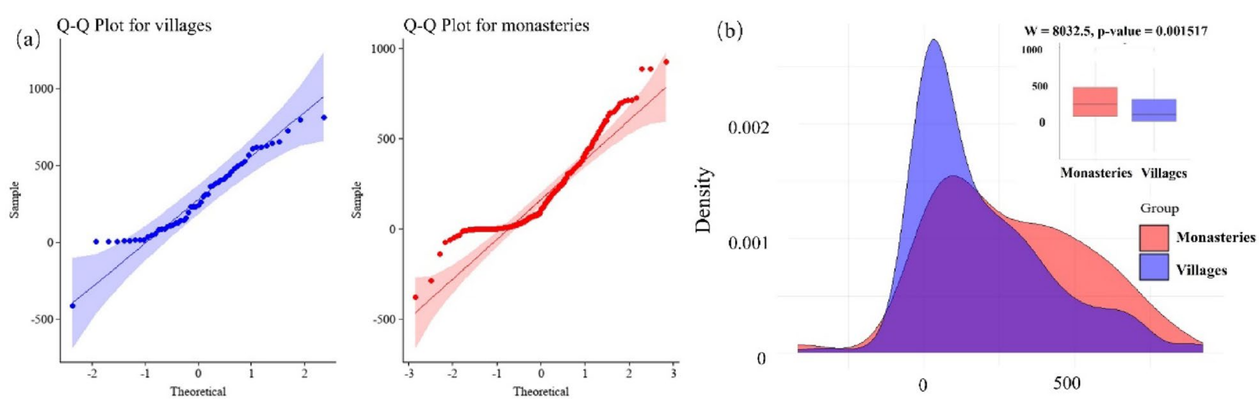


**Fig. 6** Significance test for differences in forest change in monastery and village buffer zones. **a** Denotes the extent to which the distribution of the data deviates from the normal distribution; **b** denotes the Mann–Whitney U test for the two sets of data

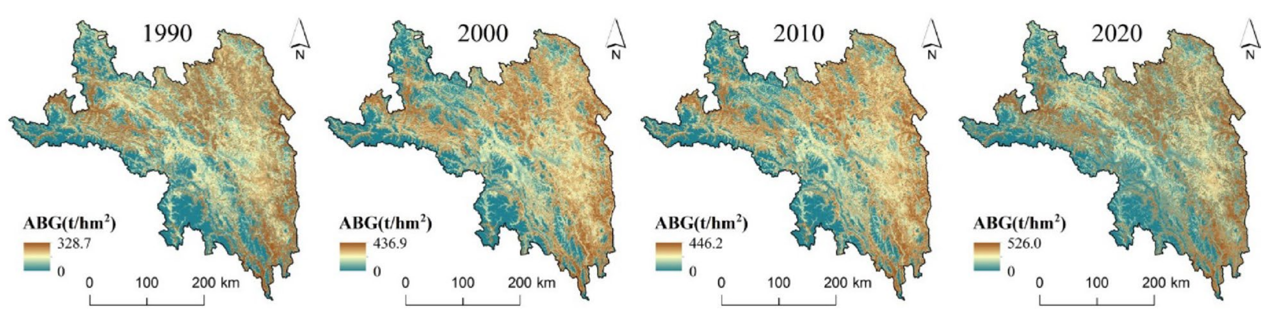


**Fig. 7** Above-ground biomass around villages and monasteries in Changdu

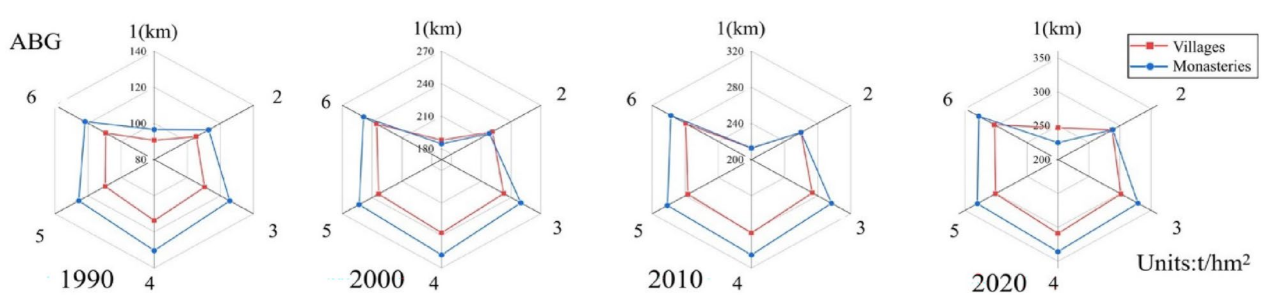


**Fig. 8** Comparison of ABG for different distance buffer zones

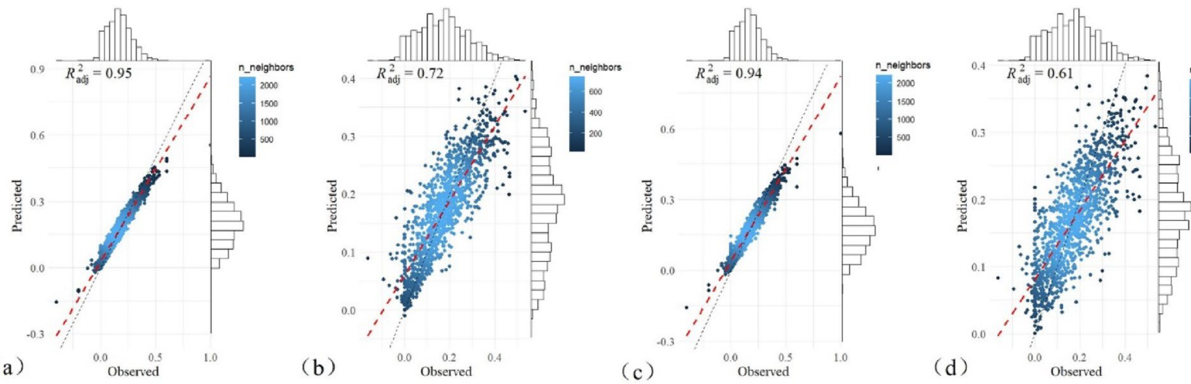

**Fig. 9** Results of random forest fitting in the buffer zones of monasteries and villages. **a**, **b** Represent the fitting results of the monastery point training set and test set, respectively; **c**, **d** represent the fitting results of the training set and test set of village points, respectively

zones (adjusted R-squared = 0.72) outperformed that within village buffer zones (adjusted R-squared = 0.61). These findings underscore the effectiveness of random forest regression in providing insights into the factors driving forest quality around these specific points of interest.

To further explore the influence of monastery sites on forest quality, we conducted a ranking of factors. The results indicate that within monastery buffer zones, the primary influence stems from initial aboveground biomass. Aboveground biomass, as a crucial component in sustaining ecosystem material cycles (Fig. 10), exerts the greatest impact on forest quality. Additionally, precipitation, elevation, and temperature, as well as monastery density, demonstrate notable effects on forest quality. Conversely, human activities including changes in population size and grazing exert relatively weaker influences on forest quality variation. In the buffer zones around village points, the importance of initial aboveground biomass diminishes, with climatic factors (precipitation and temperature) and topographical factors (elevation and slope) predominantly shaping forest quality variation in this area. Human activities exhibit an increased influence on forest quality, particularly regarding grazing intensity, although their overall impact remains limited. By comparing the driving forces behind forest aboveground biomass in monastery areas and villages, we found that climate change is the predominant factor influencing biomass variation at a regional scale. However, local efforts to implement effective measures in response to broader

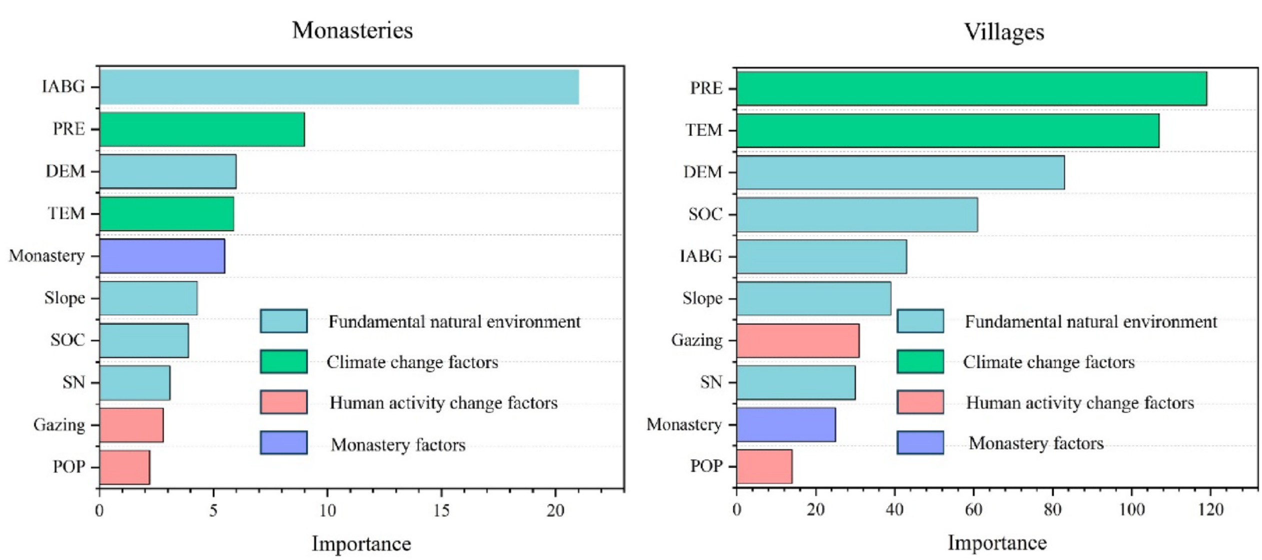


**Fig. 10** Importance of factors influencing forest quality around monastery and village sites

climate change challenges remain difficult. In contrast, the protective role of monasteries on forest biomass exceeds the environmental impacts of population growth and grazing. Therefore, monasteries have become a crucial mechanism for effective forest conservation.

After investigating the significance of various driving factors, we were constrained by the limitations of the random forest model, which prevented us from definitively determining the positive or negative influence of each driver on forest quality. The diagonal shows univariate partial dependence plots (illustrating the effect of single variables on the target), while above are bivariate plots (showing interactions between variables), and below are scatter plots of the raw data for each variable. Consequently, we turned to partial dependence relationships to gain insights into the impact of the primary driving factors within the monastery buffer zone on the dependent variable. Our findings revealed that the initial aboveground biomass did not exhibit a significant trend. It only showed a growth trend within a lower range of aboveground biomass (Fig. 11). The partial dependence relationship for the DEM factor indicated that an increase in altitude within the [3000–4000 m] range corresponded with an improvement in forest quality. Simultaneously, we discovered a significant positive correlation between precipitation and monastery factors influencing forest quality. This suggests that areas with increased rainfall and closer proximity to monasteries often exhibit higher positive changes in forest quality.

## Analysis of impact pathways

To analyze the pathways of various factors affecting forest quality, we employed PLS-SEM to fit the independent and dependent variables. The Goodness of Fit (GOF) for all models exceeded 0.36 (Table 1), indicating strong overall fit of the model. The primary driving

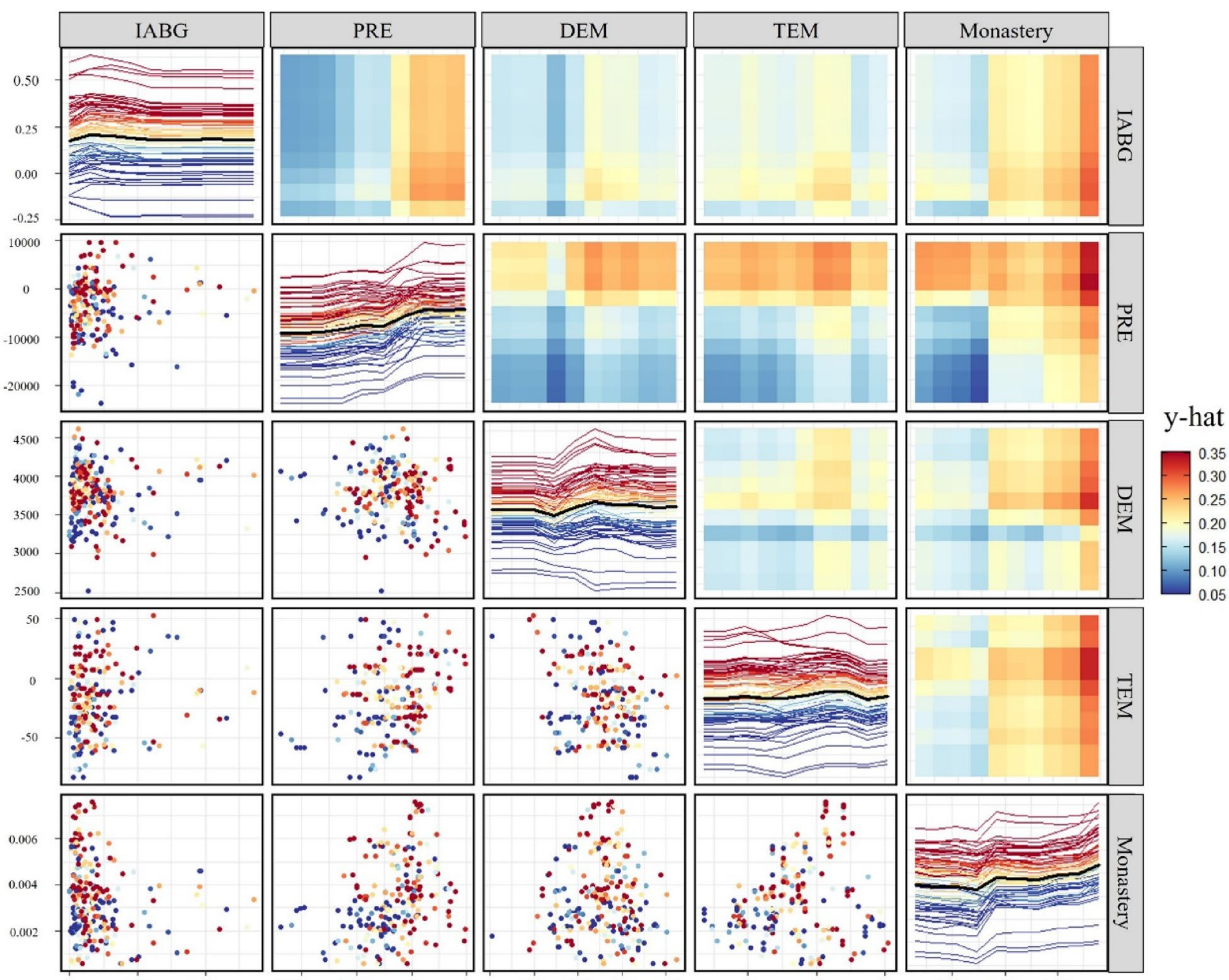


**Fig. 11** Partial dependence of the main drivers within the monastery buffer zones

**Table 1** PLS-SEM model fitting results

| Path | Villages | | | Monasteries | | |
|---|---|---|---|---|---|---|
| | Effect | p-value | GOF | Effect | p-value | GOF |
| Climate change → ABG | 0.7748 | 0.000 | 0.396 | 0.4948 | 0.000 | 0.432 |
| Fundamental natural environment → ABG | 0.359 | 0.000 | | 0.3584 | 0.000 | |
| Human activity change → ABG | −0.1259 | 0.008 | | −0.0816 | 0.090 | |
| Monastery → ABG | 0.0937 | 0.070 | | 0.1484 | 0.000 | |
| Monastery → human activity change | 0.0904 | 0.000 | | 0.016 | 0.120 | |

force behind forest aboveground biomass around both monastery and village sites is climate change, showing a positive effect with impact coefficients of 0.49 and 0.77, respectively (Fig. 12). However, there is a clear difference between the effects of human activities and monastery factors on forest aboveground biomass. The negative effect of changes in human activities on forest aboveground biomass around villages is more significant, with an impact coefficient of −0.13. In contrast, the positive effect of monastery factors on the forest around monasteries is more pronounced, with an impact coefficient of 0.15. Although the monastery's effect may not be statistically significant in some areas, it generally exerts a beneficial impact on forest aboveground biomass. This indicates that, in the context of climate change, the protection of forest aboveground biomass by monasteries extends beyond just the immediate area around the monastery. It also influences the intensity of human activities, further contributing to forest conservation.

## Discussion

### Tibetan Buddhism's reverence for forest conservation

In this study, we quantitatively assessed the influence of Buddhist monasteries on forest aboveground biomass (ABG) using spatial analysis and statistical modeling techniques. By employing Random Forest regression and Partial Least Squares Structural Equation Modeling (PLS-SEM), we identified the specific drivers of forest quality and elucidated the pathways through which monasteries influence forest conservation. This approach provides empirical evidence of the monasteries' role in enhancing forest biomass, bridging the gap between qualitative cultural insights and quantitative ecological data. Our study revealed a significant increase in forest aboveground biomass (ABG) in Changdu, a culturally important region of the QTP, from 1990 to 2020, with more pronounced growth around monasteries compared to villages. This trend aligns with findings from other regions on the QTP. Sajadi et al. [45] observed a general increase in forest cover in the eastern QTP. Similarly, Liang et al. [46] reported that climate change and reforestation policies

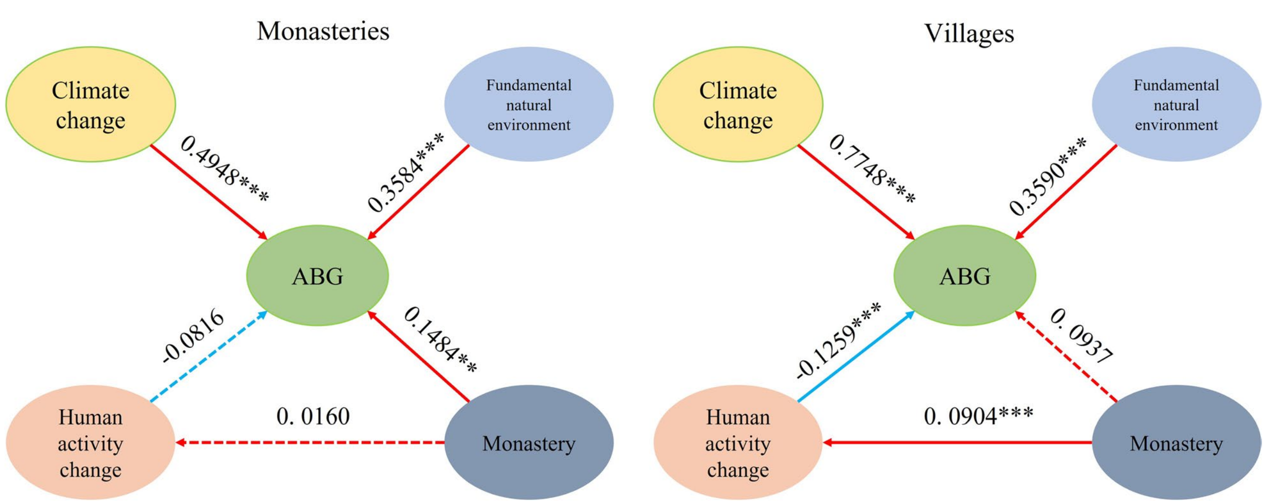


**Fig. 12** Path analysis of monastery and village buffer zones: solid lines represent significant relationships, dotted lines represent non-significant relationships

have contributed to forest expansion in the Tibetan Plateau. These studies support our observation that forest ecosystems in the QTP are experiencing positive changes, influenced by both environmental factors and human interventions.

In Tibetan Buddhism, trees are considered sacred symbols. For instance, in the *Shui Mu Ge Yan*, it is stated that: “The labor of cutting trees brings momentary exhaustion, but the torturers in hell will never tire.” In our interviews with monks and local residents, we found that monks are extremely cautious about cutting down trees. They believe that trees symbolize blessings and should be treated with deep reverence. For instance, during monastery construction, if a tree must be cut down, monks ensure that another tree is planted in its place. This practice is rooted in the concept of Karma—every action, such as cutting a tree, requires a positive counterbalance, like planting a new tree, to avoid negative Karma. The monastery, as a sacred institution, commands deep respect from the local community. Residents often mention that “wherever the sound of the monastery bell can be heard, no tree may be cut, nor bird hunted.” Monks’ teachings also encourage forest conservation among the community, with locals believing that planting and protecting trees bring blessings to their families. One resident shared that “each tree planted adds more blessings for the family.” Trees are cherished as valuable parts of their environment, and when asked about the possibility of cutting them, locals expressed that “the saplings planted in the courtyard become part of the family, and everyone takes great care of them. Despite such efforts, most saplings struggle to survive due to the harsh conditions of the plateau, where cold, low-oxygen, and unpredictable weather make these fragile green lives akin to leaves drifting in the Lancang River—quickly disappearing if not carefully tended.” When cutting a tree becomes unavoidable for house construction, locals kneel and pray, explaining their reasons to the gods and asking for forgiveness for their actions. We also found support for the positive influence of Buddhist monasteries on forest conservation in earlier studies. Allendorf et al. [47] found that sacred forests managed by Buddhist monasteries in Myanmar exhibited higher levels of biodiversity and forest cover compared to non-sacred areas. In the Himalayas, Brandt et al. [26] reported that religious beliefs and practices associated with monasteries contributed to the preservation of sacred groves, which serve as refuges for endemic species. These studies highlight the positive influence of monasteries on forest conservation, consistent with our findings that monasteries not only protect forests in their immediate vicinity but also impact broader human activities related to forest conservation.

#### Monasteries are not far from villages

The proximity of monasteries to villages is common across various cultures, especially in regions where Buddhism has a strong presence. This unique spatial arrangement raises intriguing questions about the interplay between religious beliefs, environmental conservation, and community dynamics. Monasteries often serving as centers of communal activity, spiritual guidance, and social cohesion. They offer people spaces for meditation, rituals, and religious education. In return, the local community provides crucial support to maintain and sustain these religious institutions through donations, both monetary and in-kind, which are vital for monastery upkeep and expansion. Moreover, the influx of pilgrims and tourists to monasteries stimulates local economies [6] (tourism revenue in Tibet grew from 630 million RMB in 2000 to 16.52 billion RMB in 2013, and the number of tourists rose from 564,000 in 2000 to 12.91 million in 2013), indirectly benefiting villagers’ welfare. Although tourism around monasteries can boost the local economy, rapid and unmanaged expansion threatens the sustainability of natural resources [7]. The relationship between monasteries and villages represents a dynamic interplay between religious beliefs, environmental conservation, and socio-economic interactions. Future research should continue to explore the nuances of this relationship, accounting for varying cultural contexts and evolving societal dynamics.

#### Scale effects and spatial heterogeneity as driving factors

In our study, conducted at a regional scale in Changdu, multiple linear regression analyses revealed that the coefficient of determination ($R^2$) was only 0.33, indicating that the included predictors explained only a modest portion of the variance in aboveground biomass, which was the dependent variable in the analysis. Notably, precipitation was the only variable that showed a significant correlation globally (Table 2). This highlights the challenges of capturing complex environmental interactions at varied scales.

Furthermore, when analyzing the influence of different drivers within buffer zones around monasteries and villages, distinct variations in influential factors were evident. Such differential responses underscore the necessity of adopting a localized approach to understanding these dynamics. Based on existing research on village and monastery interactions and human proximity distances, we established a 6 km buffer zone as the optimal scale for quantifying the drivers at these locations [42].

**Table 2** The results of multiple linear regression at the regional scale in Changdu

| Factors | Estimate | Std.Error | t.value | p-value |
|---|---|---|---|---|
| (Intercept) | −1.1E−01 | 2.2E−01 | −4.7E−01 | 0.64 |
| DEM | 4.9E−05 | 6.1E−05 | 8.1E−01 | 0.42 |
| Slope | 2.0E−03 | 2.8E−03 | 7.0E−01 | 0.48 |
| PRE | 1.1E−05 | 3.3E−06 | −3.3E+00 | 0.00 |
| TEM | 6.7E−04 | 6.4E−04 | 1.0E+00 | 0.29 |
| SOC | 1.0E−03 | 1.5E−02 | 6.7E−02 | 0.95 |
| SN | 2.7E−01 | 4.1E−01 | 6.7E−01 | 0.51 |
| POP | 4.7E−03 | 1.7E−03 | 2.7E+00 | 0.63 |
| Gazing | −2.2E−04 | 6.9E−05 | −3.2E+00 | 0.82 |
| IABG | −1.8E−05 | 1.9E−05 | −9.6E−01 | 0.34 |
| Monastery | 2.4E+01 | 1.1E+01 | 2.1E+00 | 0.24 |

Simultaneously, we acknowledge significant spatial heterogeneity among various driving factors. Particularly, the intensity of human activities in Changdu has shown pronounced regional variations. Over the period from 1990 to 2020, human activity index (HAI) in Changdu has steadily increased, with average intensity values rising from 0.017 to 0.057 (Fig. 13). Initially concentrated in the eastern region, significant shifts in activity concentration were observed moving to the northern area by 2000 and later expanding to the southeastern region by 2020. These shifts in human activity patterns not only reflect the dynamic nature of regional development but also highlight the spatial variability in environmental and social drivers affecting Changdu. This spatial heterogeneity necessitates tailored management strategies that can effectively address the localized impacts of these driving factors.

## Study limitations and future research

The Random Forest algorithm offers several analytical advantages, including the ability to model complex, non-linear relationships and handle high-dimensional data without overfitting. It provides variable importance measures, helping to identify key drivers affecting forest quality. However, one limitation is that it operates

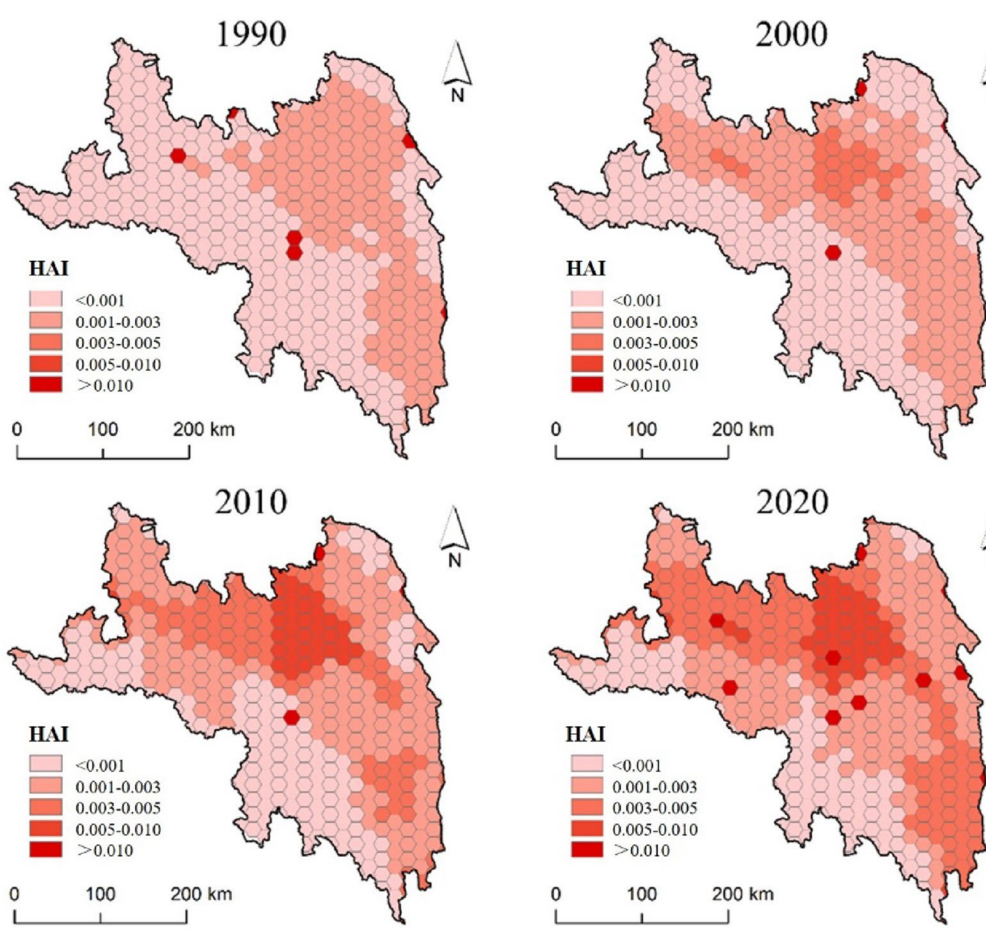


**Fig. 13** Changes of the human activity index in Changdu from 1990 to 2020

as a "black box," making it challenging to interpret the specific nature of the relationships between variables [4]. Additionally, data accuracy plays a crucial role in assessing forest conditions, requiring high-resolution data over a long time series to precisely capture changes in forest ecosystems [48]. In this study, we focused solely on quantifying the impact of monasteries on forest biomass in Changdu. However, the practical involvement of monasteries in forest conservation efforts remains underexplored. To address this gap, future research should involve selecting representative monasteries to evaluate their long-term contributions to forest conservation practices. This includes assessing the influence on residents' perceptions as well as the mediating role that monasteries may play between local communities and government management.

## Conclusion

This study provides a comprehensive analysis of the role that Tibetan Buddhist monasteries play in forest conservation on the Qinghai-Tibet Plateau, with a focus on the Changdu region. Key findings include: (1) Spatial clustering of monasteries and villages: the study reveals significant spatial clustering of Tibetan Buddhist monasteries and nearby villages. This clustering is not random but shows a structured relationship, indicating a strong spatial interaction between monasteries and villages. (2) Positive influence on forest conservation: forests near monastery sites exhibit higher aboveground biomass and a more significant increase in forest area compared to areas around villages. Monasteries positively influence forest conservation through a combination of religious practices and community engagement. Monks protect forests near monasteries by carefully managing tree cutting and replanting, guided by religious beliefs such as Karma. Additionally, monasteries shape the environmental awareness of surrounding communities, encouraging them to adopt sustainable practices and protect local forests through coordinated efforts. (3) Drivers of forest quality: utilizing random forest and partial least squares structural equation modeling (PLS-SEM), the study identifies the primary drivers of forest quality. Initial aboveground biomass, climatic factors (precipitation and temperature), and monastery density emerge as key influences on forest quality within monastery buffer zones. (4) Spatial heterogeneity of driving pathways: the pathway analysis underscores the spatial heterogeneity of driving factors affecting forest quality. The density of monasteries within buffer zones plays a significant role in forest protection, indicating that cultural factors are crucial in shaping forest landscapes. Conversely, the areas surrounding villages do not significantly contribute to the protection of forest quality, as these areas are primarily influenced by the negative impacts of human activities.

### Abbreviations

| | |
|---|---|
| QTP | Qinghai-Tibet Plateau |
| SEEA-EA | The System of Environmental-Economic Accounting Ecosystem Accounting |
| PRE | Average annual precipitation |
| TEM | Average annual temperature |
| IABG | Initial aboveground biomass |
| DEM | Digital elevation model |
| SOC | Soil organic carbon |
| SN | Soil total nitrogen |
| POP | Population |
| ABG | Aboveground biomass |
| GOF | Goodness of fit |
| HAI | Human activity index |

### Author contributions

Conceptualization, S.Z., and L.G.; formal analysis, S.Z., and T.W.; funding acquisition, L.G.; investigation, S.Z; methodology, S.Z and L.G.; project administration, L.G.; resources, S.Z, T.W. and L.G.; writing—original draft, S.Z.; writing—review and editing, T.W and L.G. All authors have read and agreed to the published version of the manuscript.

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### Data availability

No datasets were generated or analysed during the current study.

## Declarations

### Competing interests

The authors declare no competing interests.

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## Publisher's Note